

HW 3

```
#load libraries
library(stats)
library(ggplot2)
library(DescTools)
```

Mode function which will be used through the HW.

```
Mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
```

1a. Let U follow $\text{Uniform}(a,b)$ Use simulations to numerically estimate the median and the mode of U for $a = 0$ and $b = 2$.

```
#set seed
set.seed(99)

uniform_values = runif(100, min = 0, max = 2)

median(uniform_values)
```

```
## [1] 1.005558
```

```
Mode(uniform_values)
```

```
## [1] 1.169424
```

2a. Use R to numerically estimate the median and the mode of $X \sim \text{Expo}(2)$

```
set.seed(99)

expo_values = rexp(100, rate = 2)

median(expo_values)
```

```
## [1] 0.3119728
```

```
Mode(expo_values)
```

```
## [1] 0.08471185
```

3a. Use R to numerically estimate all means and modes of X for $n = 1, 2, \dots, 10$.

```
set.seed(99)  
mean(runif(1))
```

```
## [1] 0.5847119
```

```
Mode(runif(1))
```

```
## [1] 0.1137817
```

```
mean(runif(2))
```

```
## [1] 0.8383868
```

```
Mode(runif(2))
```

```
## [1] 0.5349936
```

```
mean(runif(3))
```

```
## [1] 0.4414561
```

```
Mode(runif(3))
```

```
## [1] 0.1753148
```

```
mean(runif(4))
```

```
## [1] 0.5396829
```

```
Mode(runif(4))
```

```
## [1] 0.3578854
```

```
mean(runif(5))
```

```
## [1] 0.4989098
```

```
Mode(runif(5))
```

```
## [1] 0.350086
```

```
mean(runif(6))
```

```
## [1] 0.3667029
```

```
Mode(runif(6))
```

```
## [1] 0.6909805
```

```
mean(runif(7))
```

```
## [1] 0.6396811
```

```
Mode(runif(7))
```

```
## [1] 0.1102899
```

```
mean(runif(8))
```

```
## [1] 0.5691436
```

```
Mode(runif(8))
```

```
## [1] 0.4418447
```

```
mean(runif(9))
```

```
## [1] 0.4454068
```

```
Mode(runif(9))
```

```
## [1] 0.647061
```

```
mean(runif(10))
```

```
## [1] 0.3899278
```

```
Mode(runif(10))
```

```
## [1] 0.05744927
```

6a. Let X and Y be iid $\sim \text{Unif}(0,1)$. Numerically estimate the covariance of $X+Y$ and $X-Y$.

```
#we know covariance is E(XY) - E(X)E(Y)
```

```
set.seed(100)
```

```
x = runif(10000, min = 0, max = 1)
```

```
y = runif(10000, min = 0, max = 1)
```

```
cov(x+y, x-y)
```

```
## [1] 0.0003780625
```